

2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-3P

(Data Structure)

Full Marks : 25

SET - II

Marks Distribution :

Experiment : 15

Source Code : 08

Algorithm : 04

Output : 03

Viva voce : 05

Laboratory Notebook : 05

Answer *any one* question.

1. Input a list of numbers in arbitrary order and construct a priority queue using Max Heap. Write a program in C to sort this list of numbers utilizing Max Heap.
2. Write a program in C that takes as input a binary tree and traverse it using post-order traversal.
3. Input a list of numbers in arbitrary order and write a program in C to construct binary search tree from these numbers. Also write functions for insertion and deletion of elements from this constructed binary search tree.

4. Write a C program that checks whether a string of parentheses is balanced or not using stack.
Input an expression in parentheses : {[}]
Expected Output : The expression is not balanced.
Input an expression in parentheses : ((()))
Expected Output : The expression is balanced.
5. Write a C program to calculate the height of a binary tree. Ensure the program handles empty trees gracefully.
6. Write a C program to delete the n-th element of a circular queue.
Firstly, this code checks if the queue is empty or if the position to delete is invalid. If the position is valid, it deletes the first (n - 1) elements, then deletes the n-th element by calling the dequeue() function.
7. Write a program in C that input a list of numbers in arbitrary order and perform sorting using Mergesort algorithm.
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